

An Introduction

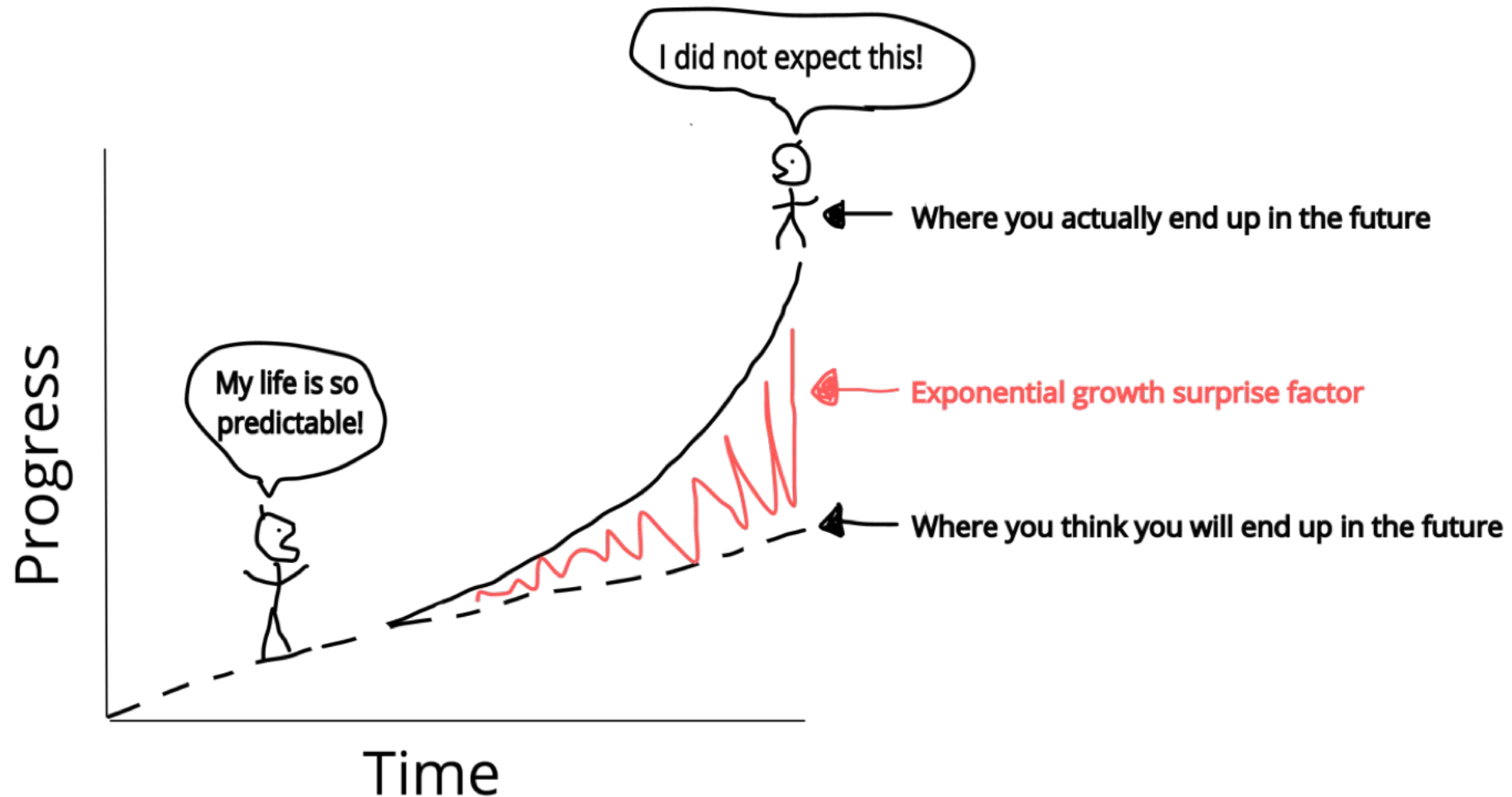
What is changing exponentially around us?



- Computing speed
- Data creation

We are living in an exponential and a global world. Something that happens in some other parts of the world affects you minutes later, through stock prices, news, viruses, etc.

Are we able to understand this growth?



Two decades back!



ASCI Red

- Commissioned: 1997
- Cost: 55 million USD
- Sponsors: Intel
- Speed: 1.3 Teraflops

One decade back!



Sony PS 3

- Year: 2006
- Price: 499 USD
- Speed: 2.1 Teraflops

What is driving this change?



- Data
- Computing
- Development of learning algorithms

What is machine learning?



- Data mining: Helps in discovering new knowledge from large data bases
- Machine learning: Create systems with the help of data that can solve specialized problems like a human or even better than a human
- Artificial intelligence: Any artificial system that is non-living but shows some form of intelligence

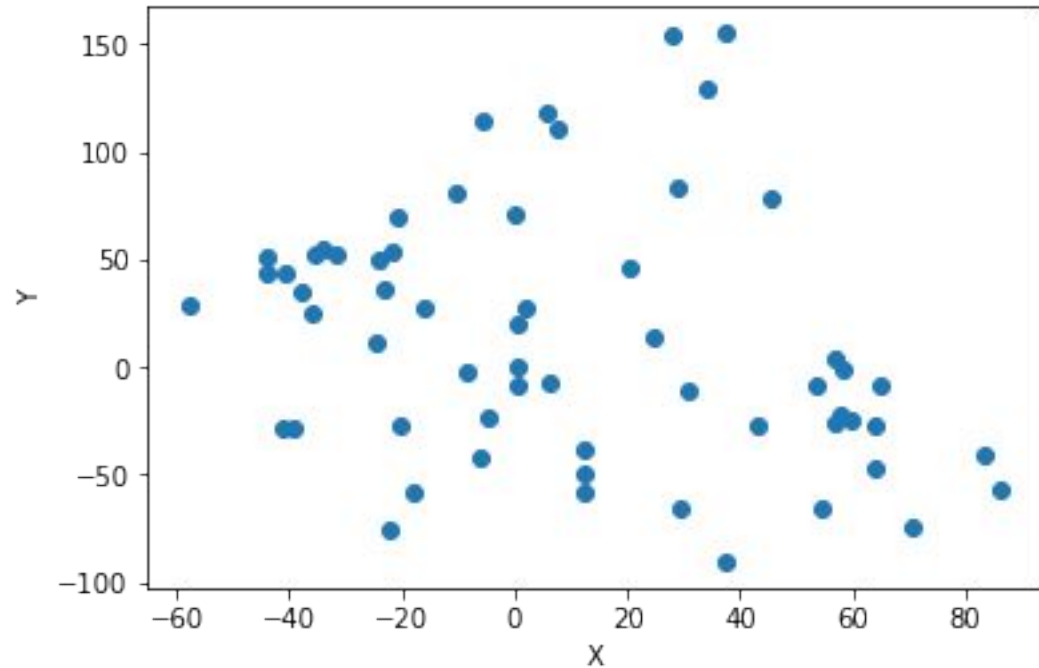
- Support vector machines, decision trees



Loan acceptance dataset

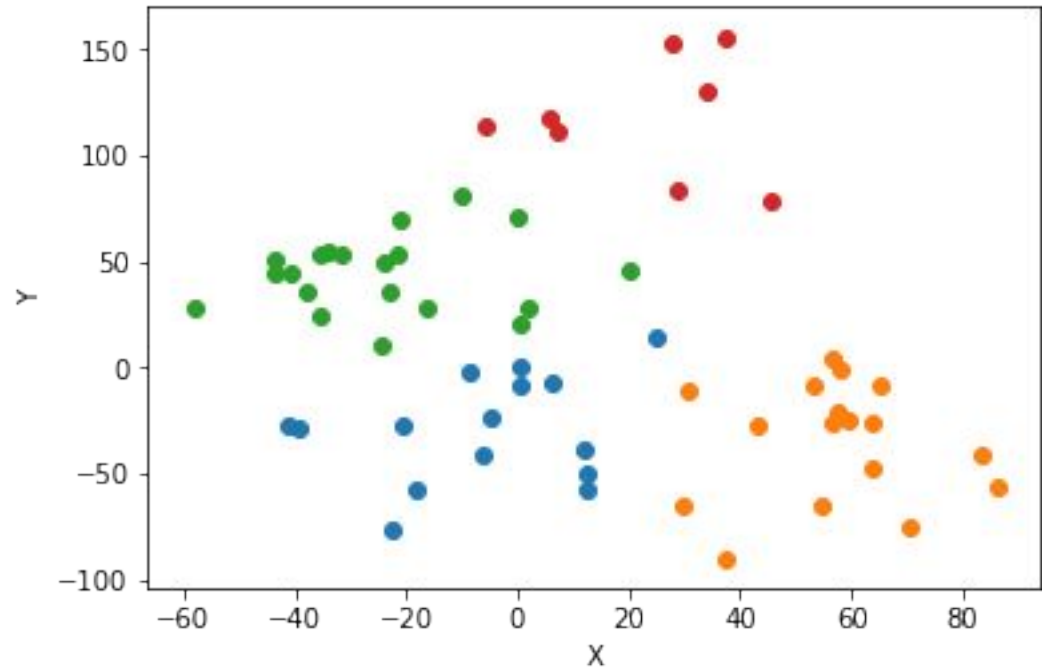
	Marital Status	Kids	Annual Household Salary	Loan Amount	Car owner	Education Level	Predict Loan Granted
1							
2	0	0	106020	185913	1	0	0
3	0	2	1270279	4907598	1	0	0
4	1	3	1086365	3012855	1	0	1
5	1	2	516564	2493950	1	0	0
6	0	2	1291768	4870552	1	5	1
7	0	1	813425	1815248	0	0	0
8	1	0	940121	2039002	1	5	1
9	1	0	1211396	1373174	1	4	1
10	0	1	1508588	5926431	1	4	1
11	1	2	4458464	24451446	0	5	1
12	0	2	607875	1844438	0	0	0
13	0	0	477131	1372348	1	4	1
14	1	2	339217	918574	1	1	1
15	1	0	1183169	4281420	0	1	1
16	0	2	1455947	6360417	1	0	0
17	1	1	258602	1260440	1	0	1

- Clustering, optimization



- Clustering, optimization

E-Commerce
Logistics
Industry



- Neural Networks



Handwriting recognition

- Deep learning



Object detection

•Natural language

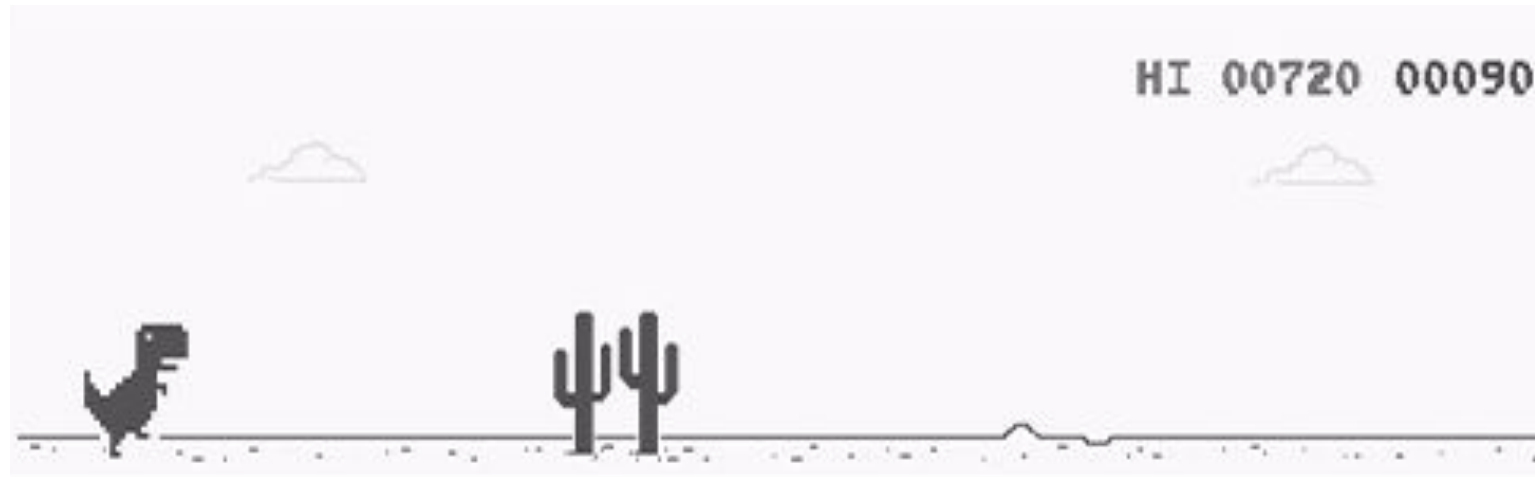
processing

S No.	Title	Decisions
1	SpiceJet to issue 6.4 crore warrants to promoters	{'SpiceJet': 'hold'}
2	Mid-cap funds can deliver more, stay put: Experts	{'Mid-cap funds': 'buy'}
3	Mid caps now turn into market darlings	{'Mid caps': 'buy'}
4	Market seeing patience, if not conviction: Prakash Diwan	{'Market': 'hold'}
5	Infosys: Will the strong volume growth sustain?	{'Infosys': 'hold'}
6	Hudco raises Rs 279 cr via tax-free bonds	{'Hudco': 'buy'}
7	HOEC could retest 30-35 levels: Ashwani Gujral	{'HOEC': 'hold'}
8	Gold shines on seasonal demand; Silver dull	{'Gold': 'buy', 'Silver': 'sell'}
9	Genpact appoints Edward J Fitzpatrick as CFO	{'Genpact': 'hold'}
10	EXL beats profit estimates, cuts sales outlook	{'EXL': 'buy'}
11	Wait and watch on Bharti Airtel: Vinay Khattar	{'Bharti Airtel': 'hold'}
12	Would stick to banking: Girish Pai, Centrum Broking	{'banking': 'buy', 'Centrum Broking': 'hold'}
13	MSCI adds Aurobindo Pharma to India index	{'Aurobindo Pharma': 'buy', 'MSCI': 'hold', 'India index': 'hold'}
14	Ashok Leyland raises Rs 667 crore through QIP	{'Ashok Leyland': 'buy'}
15	At Wipro, growth remains a mirage	{'Wipro': 'sell'}

Financial News Analysis

Buy, Sell or Hold stocks based on news

- Computer vision



Make T-Rex, the dinosaur jump with gestures

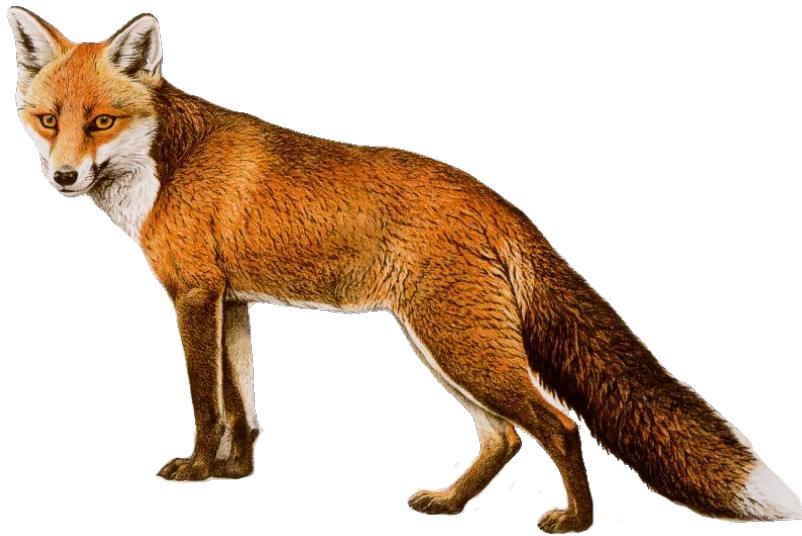
Supervised vs Unsupervised Learning

Supervised Learning

Learns based on
instruction

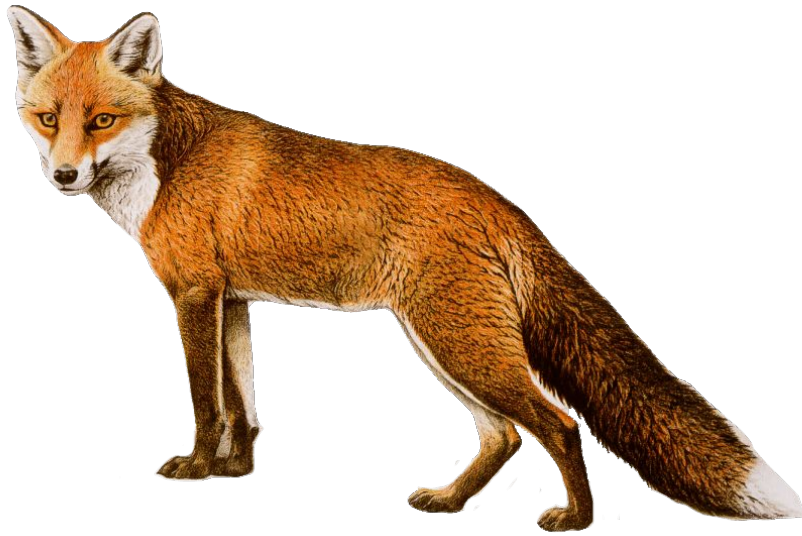


Supervised learning: What about a fox?



The child will call it a dog.

Supervised learning: Model update!



The child will call it a dog.

New data requires model update.

Unsupervised Learning

Learns based on similarity



Learns based on size and shape of the face.

Learns based on similarity



Small dog with flat face.

Mogli thinks that it is a cat!

Unsupervised Learning: Model update!



Small dog with flat face.

Mogli thinks that it is a cat!



Over time adds one more dimension to its model ☐ Sound

Transfer Learning



A sadhu living in Indian forests recognizes monkeys and wolves



Transfer Learning



The sadhu goes on a trip to African forests where he sees baboons and hyenas

